Label Consistency and Iconicity in Tactile Category Learning

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Linguistic labels shape visual categorization, but their role in tactile perception, a sensory modality difficult to verbalize, remains unexplored. Japanese mimetic words (e.g., *zarazara* for rough textures) offer iconic sound-meaning correspondences (Imai & Kita, 2014; Akita, 2009), unlike arbitrary labels used in visual studies. If labels provide feedback to perceptual systems, enhancing discrimination within labeled categories (Lupyan, 2012), label-percept consistency should be critical for tactile category learning. We investigate whether consistent tactile-linguistic pairings facilitate learning more than inconsistent pairings in native Japanese speakers. Building on perceptual learning research (Goldstone, 1998; Vanek et al., 2019), we hypothesize that (H1) consistent label-texture pairings produce larger learning gains than inconsistent pairings, with (H2) effects persisting without labels; indicating genuine category formation rather than label-dependent associations.

52 Japanese speakers (N=26 per condition) completed 4-day tactile categorization training using ABX triplet discrimination (Vanek et al., 2021). Participants discriminated 18 fabric types organized into 6 triplets (Hanada, 2023) based on friction, softness, and thickness, paired with auditory labels. Verbal stimuli comprised three types (see Table 1): JP-Match, JP-Pseudo, and Eng-Pseudo (Rastle & Coltheart, 2000). In Concordant condition, 75% of pairings used JP-Match aligned with perceptual structure, with 25% using misaligned pseudowords. Discordant condition reversed this ratio. Training comprised 4 blocks (6 triplets per block) per day over 4 days, with immediate feedback (Figure 1). Following Day 4 training, participants completed an unlabeled test using Day 1 triplets to measure learning independent of linguistic cues. We recorded accuracy, reaction times, and confidence (9-point scale). Data were analyzed using GLMMs (Figures 2-3 detail model specifications and statistics), and the confidence-learning correlations were computed per condition.

Concordant training produced substantially larger learning gains than Discordant training (Figure 2A; d=0.96, p=.001), with Concordant participants maintaining 86% accuracy at the test. The condition × session interaction was significant (Figure 2C; p<.001), showing consistent labels amplified learning by 54%. Concordant participants responded 17% faster than Discordant participants (Figure 3A; d=-2.37, p<.001). This speed advantage held for both correct and incorrect responses (Figure 3B; interaction p=.14), ruling out speed-accuracy trade-offs. Confidence ratings increased in Discordant training despite minimal learning gains (Δ=+0.65 vs -0.07, p<.001), with confidence improvements correlating with learning only in Concordant training (r=.37, p=.059 vs r=.11, p=.60).The obtained results confirm both hypotheses: consistent tactile-linguistic pairings produced larger learning gains than inconsistent pairings (H1), with effects persisting at the test session (H2), demonstrating actual category formation rather than label-dependent associations. The persistence of effects without linguistic cues shows that consistent labels enable perceptual reorganization rather than mere associative learning, which supports the Label Feedback Hypothesis in the tactile domain.

These effects emerged in native Japanese speakers using their mimetic vocabulary, showing that label consistency shapes tactile perception even when speakers possess rich sensory-linguistic representations. This occurred despite participants' established sound-symbolic knowledge: rather than facilitating learning, existing iconicity knowledge appears to have created interference when labels misaligned with perceptual structure, suggesting label-percept consistency matters more than iconicity per se. That native speakers showed such robust sensitivity reveals that linguistic structure imposes fundamental constraints on perceptual category formation across sensory modalities.

Figures and Graphs

Table 1: Verbal stimulus types and examples

Type	Structure	Examples
JP-Match (Authentic /Target)	AB-AB Sound-symbolic, mimetic	Zarazara, Mokomoko
JP-Pseudo	AB-AB Pseudowords Non-sound symbolic, Japanese mimetic-like	Lomolomo, Sunosuno
ENG-Pseudo	Bisyllabic pseudowords, arbitrary	Hogger, Dastude

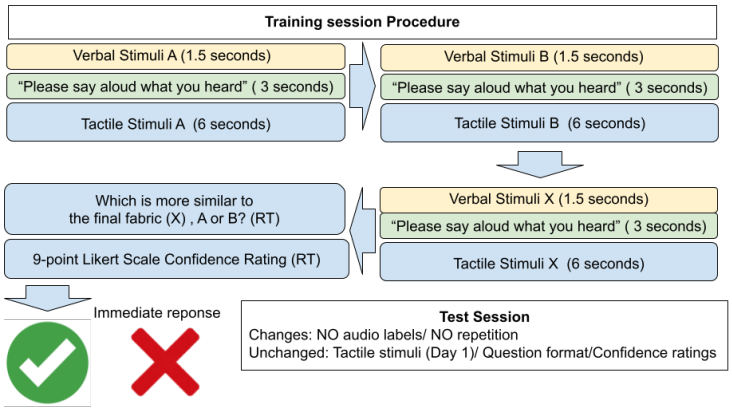

Figure 1: Experimental procedure

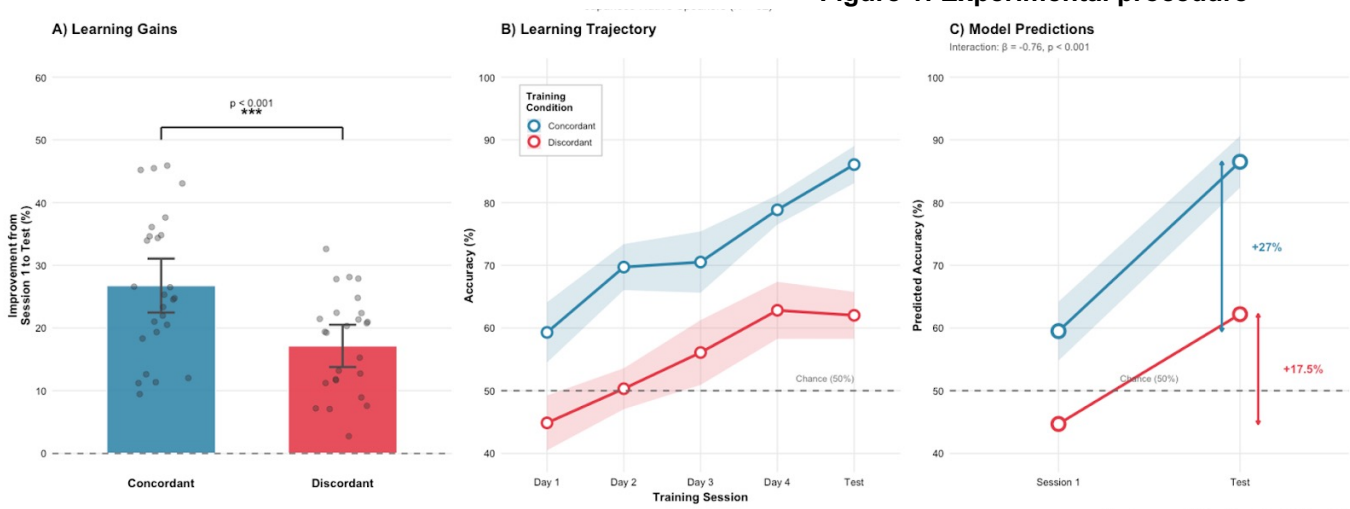

Figure 2: Effect of label consistency on tactile category learning (N=52). **(A)** Learning gains: Concordant +26.8% vs. Discordant +17.1%; $t(50)=3.45$, $p=.001$, $d=0.96$. **(B)** Accuracy trajectory across sessions; dashed line=chance. **(C)** GLMM predictions: condition × session β=-0.76, $p<.001$. **Model: accuracy ~ condition × session_type + (1|participant) + (1|triplet_id).**

Figure 3: Reaction time analysis. **(A)** Decision times: Concordant M=4.78s vs. Discordant M=5.59s; $t(50)=-8.54$, $p<.001$, $d=-2.37$. **(B)** No speed-accuracy tradeoff: condition × accuracy interaction $p=.14$ (n.s.). **Model: rt_decision ~ condition × accuracy + (1|participant) + (1|triplet_id).**

Selective References: Akita, K. (2009). A grammar of sound-symbolic words in Japanese. Doctoral dissertation, Kobe University.;Hanada, M. (2023). Dimensional structure of tactile textures characterized by Japanese onomatopoeia. *Attention, Perception, & Psychophysics*, 85, 2669-2680.; Imai, M., & Kita, S. (2014). The sound symbolism bootstrapping hypothesis for language acquisition and evolution. *Philosophical Transactions of the Royal Society B*, 369(1651), 20130298.; Lupyan, G. (2012). Linguistically modulated perception and cognition: The label-feedback hypothesis. *Frontiers in Psychology*, 3, 54.; Vanek, N., Sóskuthy, M., & Majid, A. (2021). Consistent verbal labels promote odor category learning. *Cognition*, 206, 104485.